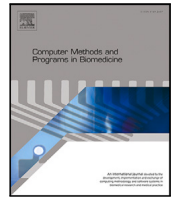




Contents lists available at ScienceDirect

Computer Methods and Programs in Biomedicine

journal homepage: <https://www.sciencedirect.com/journal/computer-methods-and-programs-in-biomedicine>



From segmentation to explanation: Generating textual reports from MRI with LLMs

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ARTICLE INFO

Keywords:

Explainability
Large language models
Medical imaging
Report generation
Semantic segmentation

ABSTRACT

Background and Objective: Artificial Intelligence (AI) has significantly advanced medical imaging, yet the opacity of deep learning models remains challenging, often reducing the trust of medical professionals towards AI-driven diagnoses. As a result, there is a strong focus on making AI models more transparent and interpretable to boost healthcare providers' confidence in these technologies.

Methods: This paper introduces a novel approach to enhance AI explainability in critical medical tasks by integrating state-of-the-art semantic segmentation models with atlas-based mapping and Large Language Models (LLMs) to produce comprehensive, human-readable medical reports. The proposed framework ensures that the generated outputs are factual and contextually rich. Our anti-hallucination design, which combines structured JSON with prompt constraints, is a critical innovation compared to most naïve LLM report generation methods, thereby enhancing the transparency and interpretability of AI systems.

Results: Experimental results show that the SegResNet model achieves high segmentation accuracy, while LLMs (Gemma, Llama, and Mistral) demonstrate diverse strengths in generating explanatory reports. Numerous metrics have been employed to assess the quality and effectiveness of generated textual explanations, such as lexical diversity, readability, coherence, and information coverage.

Conclusions: The method has been specifically tested for brain tumor detection in glioma, one of the most aggressive forms of cancer, and subsequently applied to multiple sclerosis lesion detection to further validate its generalizability across various medical imaging scenarios, thereby contributing to the trustworthiness of healthcare AI applications.

Reproducibility: The complete source code for implementing the framework and reproducing the results is publicly available, along with full pipeline examples demonstrating each step – from segmentation to report generation – at the following repository: <https://github.com/albertoalario/from-segmentation-to-explanation>.

1. Introduction

Artificial Intelligence (AI) is playing an increasingly prominent role in healthcare, particularly in medical imaging, where deep learning models have achieved remarkable performance in tasks such as disease detection, classification, and segmentation. However, despite these advances, the opacity of such models remains a critical barrier to clinical adoption. The inability to understand how and why a model arrives at a particular decision raises concerns among healthcare professionals, especially when diagnostic decisions can directly affect patient outcomes [1,2].

Explainable AI (XAI) has thus emerged as a crucial research area, aiming to enhance transparency and foster trust in AI-assisted diagnostics. In medical imaging, visual explanation methods like Grad-CAM [3]

have been widely used to highlight relevant regions in the input data. Nevertheless, these methods often provide limited insight: heatmaps can be ambiguous, emphasize clinically irrelevant features, or fail to articulate the rationale behind a decision in a manner that is human-understandable [4]. As a result, they may not align with the interpretive practices of medical experts.

Recent developments in Large Language Models (LLMs) offer new opportunities for generating textual explanations that resemble clinical narratives. Such models have demonstrated the ability to produce coherent and contextually rich descriptions. However, their outputs can suffer from factual inaccuracies, especially when prompts are poorly formulated or when the model lacks grounding in domain-specific data [5]. This limitation is particularly problematic in the medical

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<https://doi.org/10.1016/j.cmpb.2025.108922>

Received 28 October 2024; Received in revised form 30 May 2025; Accepted 17 June 2025

Available online 1 July 2025

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domain, where incorrect or hallucinated information can undermine trust and compromise safety.

While visual and textual explainability have often been treated as separate modalities, their integration holds the key to more comprehensive and clinically useful AI explanations. Visual cues can guide attention to relevant image regions, while text can convey diagnostic reasoning and contextual interpretations. When appropriately combined, these two modalities can complement each other, reflecting the way clinicians interpret and report findings.

This paper presents a novel framework that integrates visual explanations and LLMs to generate clinically interpretable reports from 3D brain MRI segmentations automatically. Unlike traditional approaches that rely on pre-existing text corpora or external knowledge bases, our methodology generates explanations from scratch, grounded directly in the model's outputs and anatomical context.

The framework is applied primarily to the detection of glioma tumors in adult brain MRIs, and its generalizability is further evaluated on a separate dataset focused on multiple sclerosis lesion segmentation. The process begins with the application of state-of-the-art deep learning models for the semantic segmentation of brain anomalies. The resulting segmentation maps are then enriched with anatomical context through a mapping module based on the Julich Brain Atlas, which localizes each detected region within specific brain structures.

These anatomically grounded maps are transformed into structured intermediate representations – formatted in JSON – that encode spatial and diagnostic attributes in a format interpretable by LLMs. This structured input constrains the language model's generative process, ensuring that the final textual reports remain factually consistent and clinically relevant. The generated explanations are designed to emulate the style and reasoning found in radiology reports, making the outputs directly useful for medical professionals.

To evaluate the effectiveness of this framework, we introduce an assessment strategy that combines quantitative metrics – such as readability, coherence, and lexical diversity – with domain-specific criteria related to diagnostic completeness and clinical accuracy. Through this multi-faceted evaluation, we aim to validate both the interpretability and the practical utility of the generated reports.

In doing so, this work contributes to the development of trustworthy and human-centered AI systems in medical imaging. By integrating anatomical knowledge, visual context, and controlled language generation, we offer a new solution to the challenge of explainability – one that aligns closely with the cognitive and communicative practices of clinicians.

The rest of this paper is structured as follows. Section 2 reviews related work. Section 3 detail the materials and methods used in this study, outlining the dataset, preprocessing techniques, and the innovative framework proposed for integrating visual and textual explanations. Section 4 presents the results of our experimental analysis. Section 5 concludes the paper and discusses potential future developments of our research.

2. Background and objective

Explainable AI has gained significant traction in healthcare, particularly in medical image analysis. XAI techniques aim to enhance the transparency and interpretability of deep learning models, enabling medical professionals to understand the decision-making process and build trust in AI-based systems [6].

A substantial body of research in XAI for medical imaging focused on generating visual explanations that elucidate the model's decision-making process. Common approaches include saliency maps, gradient-based methods (e.g., Grad-CAM [3]), and relevance propagation techniques (e.g., Layer-wise Relevance Propagation, LRP [7]). These methods visually represent which regions of an input image are most influential in the model's predictions.

Specifically, in brain tumor detection and segmentation from MRI data, several studies have integrated XAI techniques with deep learning models to enhance trust in AI-based computer-aided diagnostic systems. Zeineldin et al. [8] proposed the NeuroXAI framework, which combines deep neural networks with various XAI methods to generate visual explanation maps for brain tumor classification and segmentation tasks.

Similarly, different studies have also integrated XAI techniques to provide visual explanations in the context of multiple sclerosis lesion segmentation from MRI data. Sadeghibakhi et al. [9] utilized M3d-CAM to generate 3D attention maps, visually highlighting the regions in the input images that are most influential in the model's predictions.

While these visual explanation methods have proven useful, they have limitations in providing comprehensive clinical explanations. As discussed in the introductory section, heatmaps and similar visualizations often fail to provide the contextual information and pathological significance that medical professionals require for informed decision-making [4].

Recent advances in Large Language Models have shown promise in generating textual explanations in natural language for medical imaging tasks. One notable example is ChatCAD, which integrates LLMs with computer-aided diagnosis models for medical image analysis [5]. In the ChatCAD framework, trained models first process medical images to obtain outputs such as disease classification probabilities. These outputs are then translated into natural language prompts for an LLM, which generates summarized reports and diagnostic conclusions and engages in conversations about symptoms, diagnosis, and treatment.

However, the use of LLMs to generate medical reports and textual explanations has not been widely studied. One challenge is that the quality and accuracy of the generated reports depend heavily on how the input prompts are formulated. Furthermore, there is a risk that LLMs may produce inaccurate information, emphasizing the need to base their output on verified models and established medical knowledge.

Although visual and textual explanations have been explored separately, very few examples effectively integrate both approaches. RadFormer [10] is a notable exception, combining global attention mechanisms for regions of interest with local attention on bag-of-words style feature embeddings. This approach allows RadFormer to generate heatmap visualizations and textual explanations mapped to radiological lexicons for tasks like gallbladder cancer detection in ultrasound images.

Recent studies underscore the crucial role of contextual information in generating meaningful diagnostic insights. Research, such as the work on EgoCap and EgoFormer [11], explicitly demonstrates that explanatory mechanisms must be rigorously contextualized to deliver clinically substantive interpretations. This aligns with the need to integrate visual and textual explanations to enhance interpretability, ensuring that AI-generated outputs provide relevant and actionable insights for medical professionals.

Despite these advancements, a significant gap remains in research that combines state-of-the-art visual explanation techniques with textual explanation generation in medical imaging. Our work aims to address this gap by proposing an innovative approach that integrates visual and textual explanations, offering complementary information that aligns more closely with physicians' reasoning and reporting processes. A key advantage of our approach is its ability to generate clinically meaningful reports without relying on preexisting textual datasets or external knowledge bases, thereby overcoming the limitations seen in systems like ChatCAD and RadFormer, which heavily depend on curated data or large-scale knowledge repositories.

To summarize, this work introduces a novel framework that, for the first time, integrates semantic segmentation, brain atlas-based anatomical mapping, and LLMs to generate structured and interpretable medical reports in the domain of neuroimaging. A central innovation lies in the use of a structured JSON-based intermediate representation, which mediates between the segmentation output and the language

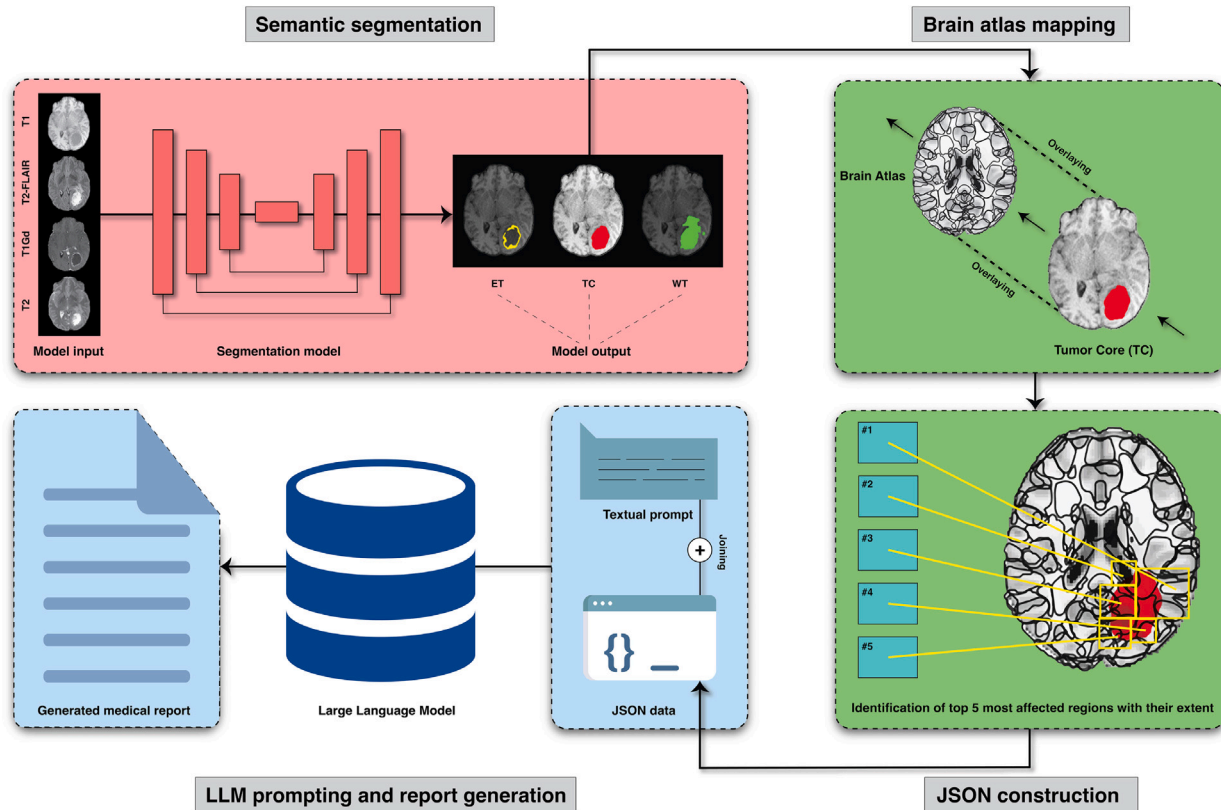


Fig. 1. Illustration of the proposed methodology pipeline, which includes semantic segmentation, brain atlas mapping, JSON construction, and LLM prompting. The workflow begins with an input MRI volume processed by the segmentation model. The resulting mask is then mapped onto the Julich Brain Atlas for anatomical region extraction. These regions are structured into a standardized JSON format, which serves as input for the prompting phase of LLMs, ultimately generating clinically interpretable textual reports.

model, ensuring factual consistency and significantly reducing the risk of hallucinations during text generation.

The framework also incorporates an error mitigation strategy that prioritizes explanations based on the most affected brain regions, thereby minimizing the influence of segmentation inaccuracies on the final report. We further present a comprehensive evaluation of the generated explanations using a range of linguistic metrics, such as coherence, lexical diversity, and readability, applied across different LLM configurations.

Building on these contributions, the following section presents a detailed description of the proposed methodology, outlining how segmentation, anatomical mapping, and language generation are seamlessly integrated to produce clinically meaningful, interpretable outputs.

3. Methods

3.1. Framework overview

We propose an innovative and explainable approach that combines advanced deep learning techniques with Large Language Models to generate interpretable medical reports in natural language. The proposed approach differs from conventional techniques in the scientific literature, as it generates textual explanations without relying on existing text related to medical images or external knowledge sources.

An essential advantage of LLMs, in fact, lies in their ability to leverage the extensive knowledge gained through pre-training on vast textual corpora. This training equips LLMs with a comprehensive understanding of medical knowledge, encompassing disease pathologies, clinical manifestations, and treatment methods. Such a rich foundation enables LLMs to effectively apply this understanding to specific medical image-processing tasks, provided they are supplied with accurate

factual data. By utilizing this rich pre-trained knowledge, LLMs can generate accurate, relevant, and contextually appropriate textual explanations that reduce hallucinations commonly associated with generative models, thereby enhancing the overall interpretability and usefulness of AI-generated insights in medical imaging and assisted diagnostics.

The proposed methodology comprises four stages, as shown in Fig. 1:

- **Semantic segmentation:** A segmentation model processes 3D brain MRI volumes to segment pathological regions such as gliomas.
- **Brain atlas mapping:** The segmented tumor regions are registered to the MNI152 standard space and mapped onto the Julich Brain Atlas to determine the most affected anatomical areas.
- **JSON construction:** The extracted spatial and quantitative information – including affected regions, tumor distribution, segmentation confidence, and model metadata – is encoded into a structured JSON file.
- **LLM prompting:** The JSON file is used to prompt advanced LLMs (Gemma, Llama, Mistral), which generate natural language medical reports describing the tumor's location, extent, and potential clinical relevance.

In this paper, our approach is applied explicitly first to brain tumors and then to multiple sclerosis lesion segmentation in 3D brain MRI. While these initial applications demonstrate promising results within the domain of brain imaging, further validation on other organs and diseases is necessary to confirm broader generalizability. Nonetheless, the methodology is inherently adaptable and holds potential for extension to other anatomical contexts, provided suitable segmentation models and structural atlases are available. The use of a detailed human body atlas enables the extraction of semantically meaningful, anatomically grounded information that can serve as factual input for LLM-based report generation.

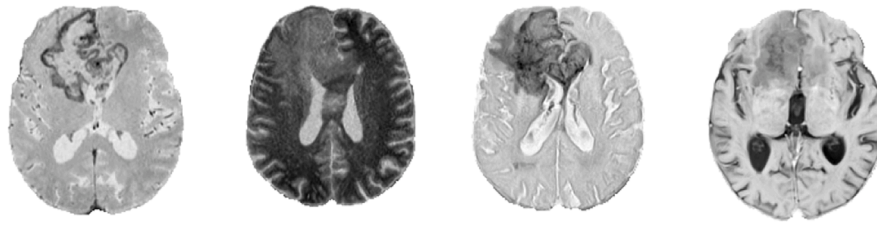


Fig. 2. MRI modalities in the BraTS 2023 dataset. From left to right: T1Gd, T1, T2-FLAIR, and T2 images. Each modality provides complementary anatomical and pathological information, contributing to a comprehensive understanding of tumor characteristics and brain tissue differentiation.

For instance, when examining liver conditions like cirrhosis or tumors, structural alterations such as nodules or masses help identify an unhealthy liver [12]. By incorporating a human body atlas, structure and tissue changes become readily identifiable and quantifiable. Additionally, many diseases have secondary effects on surrounding structures [13]. The human anatomy atlas, mapping the spatial relationships between organs, blood vessels, and nerves, aids in understanding how a localized disease can impact surrounding tissues and organs.

Beyond our current focus on glioma and MS lesions, our framework can seamlessly incorporate established structural atlases or atlas-based tools, such as 3D Slicer [14], for identifying anatomical regions across various organs. For instance, the Couinaud classification system provides a structural framework for analyzing liver tumors. At the same time, lung segmentation techniques aid in identifying abnormalities within specific lung regions, such as lobes or bronchial trees. The ability to integrate these anatomical references into our medical imaging workflows further underscores the flexibility of our approach, enhancing our framework's applicability and generalizability, making it a versatile tool in medical imaging and diagnosis across multiple scenarios.

3.2. Data

We used the dataset associated with the first task of the BraTS 2023 challenge, organized by the Radiological Society of North America (RSNA), the American Society of Neuroradiology (ASNR), and the Medical Image Computing and Computer Assisted Interventions (MICCAI) society. Specifically, the dataset comprises scans from 1470 adult patients affected by brain glioma tumors [15–19].

Each MRI scan in the BraTS 2023 dataset is provided as a NIfTI file and includes the following four MRI modalities:

- Post-contrast T1-weighted (T1Gd): T1 scan taken after a contrast agent injection.
- Native (T1): Captures the brain's natural state.
- T2 Fluid Attenuated Inversion Recovery (T2-FLAIR): Suppresses the signal from free water, making it easier to see lesions and other abnormalities.
- T2-weighted (T2): Highlights fluid-filled areas.

The imaging data underwent preprocessing steps, including co-registration to a standard anatomical template, interpolation to a uniform resolution of 1 mm³, and skull stripping [15]. An example of the MRI modalities in the BraTS 2023 dataset is shown in Fig. 2.

One to four expert raters have manually annotated all images in the dataset following a standardized annotation protocol, with annotations approved by experienced neuroradiologists. The annotations include the GD-enhancing tumor (ET), the peritumoral edematous/invaded tissue (ED), and the necrotic tumor core (NCR). The sub-regions considered for evaluation were defined as follows:

- Enhancing tumor (ET): Areas showing hyper-intensity in the T1Gd modality compared to T1 and healthy white matter in T1Gd.
- Tumor core (TC): Comprises the ET and NCR parts of the tumor, typically resected during surgery.

- Whole tumor (WT): The complete extent of the disease, including the TC and ED, typically represented by a hyper-intense signal in the FLAIR modality.

3.3. Semantic segmentation

The first stage involves preprocessing the imaging data and performing data augmentation to enhance the quality and diversity of the training dataset. Intensity normalization was applied to mitigate differences in intensity distributions across scanners and imaging protocols. The image intensities were normalized based on the mean and standard deviation of the non-zero voxel values within each channel using the formula:

$$\text{Normalized Intensity} = \frac{\text{Image Intensity} - \text{Mean}}{\text{Standard Deviation}}. \quad (1)$$

Inspired by previous approaches in the literature [20,21], we also applied data augmentation techniques, including random flipping along the three spatial axes, random spatial cropping, and random intensity scaling and shifting. These techniques increase the diversity of the training data and improve the model's generalization ability.

We explored different state-of-the-art deep learning models for semantic segmentation:

- Residual U-Net [22]: This model enhances the classic U-Net architecture by incorporating residual units in both the encoding and decoding paths. It uses parametric rectified linear unit (PReLU) activations and instance normalization. The encoding path employs residual units with varying strides for downsampling, while the decoding path uses transposed convolutions for learnable upsampling.
- SegResNet [23]: Designed for 3D medical image segmentation, SegResNet is a 3D CNN based on the residual network (ResNet) design. It features an encoder-decoder structure with four downsampling blocks in the encoder and three upsampling blocks in the decoder.
- SwinUNETR [24]: This model combines a hierarchical Swin Transformer encoder with a CNN-based decoder in a U-shaped architecture. Developed for multi-modal 3D brain tumor segmentation, it achieved top performance in the BraTS 2021 challenge. SwinUNETR processes 3D MRI data by dividing it into non-overlapping patches and employs a shifted windowing approach for efficient modeling of long-range dependencies.

To prevent data leakage, a stratified *subject-wise* splitting of the dataset was employed, ensuring data from the same patient was present in either the training or testing set but not in both [25,26].

This first stage of semantic segmentation provides an initial level of *visual explainability* in medical imaging. By highlighting the affected regions of interest (ROIs) within the MRI images, radiologists and medical professionals have a clear and intuitive explanation of tumor location, size, and morphology. This approach mitigates diagnostic uncertainty arising from complex brain structures and the often subtle differences between tumorous and healthy tissues in MRI images. A properly trained segmentation algorithm demarcates these distinctions

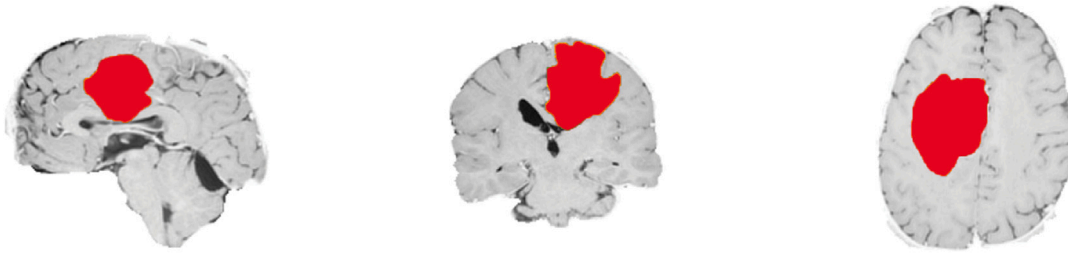


Fig. 3. Visualization of affected brain regions identified through brain atlas mapping. The mapped regions are overlaid on a 3D brain model and displayed from three anatomical perspectives: left (lateral), back (posterior), and top (superior).

by precisely outlining abnormal growths, effectively explaining areas of concern with accurate boundaries. In addition, differentiating between different tumor sub-regions with different colors clarifies which tumor parts exhibit specific characteristics. This level of detail can explain whether a tumor is growing aggressively or has spread based on the specific segmentation patterns observed.

3.4. Brain atlas mapping

In the second stage, we mapped the segmented ROIs (i.e., the tumor core) to specific brain areas using the Julich Brain Atlas [27], a unique and comprehensive three-dimensional atlas resulting as a culmination of 25 years of research. It provides detailed probabilistic maps of over 200 cortical areas and subcortical nuclei in the human brain, all based on cytoarchitectonic differences. This atlas stands out for its ability to represent the structural and functional organization of the brain, capturing inter-individual variations through the mapping of postmortem brains.

The mapping process involved overlaying the ROIs onto the standard MNI152 brain space. From there, we generated a point set representation of the tumor core voxels, which were then assigned to the corresponding brain regions in the labeled atlas map. An example of this brain atlas mapping process for brain tumor segmentation is illustrated in Fig. 3.

Through this process, we identified the top five most affected brain regions and quantified the tumor volume in each region. This stage is a significant step in understanding the impact of pathological conditions on specific brain areas.

3.5. JSON construction

We constructed a structured JSON file in the third stage to encapsulate the relevant information extracted from the semantic segmentation and brain atlas mapping stages. This step is crucial for providing the LLM with structured, factual information, thereby mitigating the risk of hallucinations—a common issue language models face when generating text without a factual basis. The JSON file serves as a source of truth, ensuring that the LLM's outputs remain consistent with the quantitative analysis and do not deviate into unreliable or fictional content.

The JSON file contains comprehensive information extracted from the previous stages for the brain tumor segmentation task. Specifically, it includes:

- The top five affected brain regions as determined during the brain atlas mapping stage, along with their respective percentages of tumor occupation and the percentage of each region affected.
- The color-coding scheme used for the semantic segmentation masks, associating specific colors with the tumor core, peritumoral edema, and GD-enhancing tumor.
- Performance metrics for the semantic segmentation, including Dice scores and Hausdorff distances for enhancing tumor, tumor core, whole tumor, and their averages.
- The name of the model used for semantic segmentation.

This structured representation ensures that the subsequent LLM prompting stage can access accurate, quantitative data derived from the image analysis, facilitating the generation of factual and trustworthy reports. The JSON template for brain tumor segmentation is shown in Listing 1.

Listing 1: Template for the JSON file given as input to the LLM in conjunction with the prompt. Capitalized strings indicate placeholders for the values.

```

1 {
2   "MRI_Scan": {
3     "Tumor_Details": {
4       "Spatial_Distribution": [
5         {
6           "Region": "REGION_AFFECTED",
7           "Percentage_of_Tumor": "%_OF_TUMOR",
8           "Percentage_of_Region_Affected": "%_OF_REGION"
9         },
10        ...
11      ],
12      "Semantic_Segmentation": {
13        "Tumor_Core": {
14          "Color": "red"
15        },
16        "Peritumoral_Edema": {
17          "Color": "yellow"
18        },
19        "GD_Enhancing_Tumor": {
20          "Color": "green"
21        }
22      },
23      "Segmentation_Confidence": {
24        "Dice_Score": {
25          "Enhancing_Tumor": "DICE_SCORE_ET",
26          "Tumor_Core": "DICE_SCORE_TC",
27          "Whole_Tumor": "DICE_SCORE_WT",
28          "Average": "DICE_SCORE_AVG"
29        },
30        ...
31      },
32      "Model_Used": "SegResNet"
33    }
34  }
35 }

```

3.6. Large language model prompting

In the final stage, we employed three state-of-the-art LLMs to generate textual explanations (i.e., medical reports) based on the information contained in the JSON file. The following LLMs were used:

- Gemma 9B-IT [28]: It is the 9 billion parameter instruction-tuned version of Google DeepMind's open-weight LLM, trained on a

Using the provided JSON data from a brain MRI study, please generate a comprehensive report detailing the segmentation results of a brain tumor. The report should be structured as follows:

1. **Summary of Tumor Segmentation:** Provide an overview of the tumor's distribution across various regions of the brain according to the Julich Brain Atlas. Discuss the semantic segmentation of the tumor into different categories (tumor core, peritumoral edema, GD-enhancing tumor) and their implications.
 2. **Detailed Regional Impact:** For each region where the tumor is present, describe the percentage of the tumor within the region and the percentage of the region affected by the tumor. Discuss the known roles and functions of these regions in the brain and how the presence of the tumor might impact these functions.
 3. **Semantic Segmentation:** Discuss the different aspects of the tumor identified by the semantic segmentation (tumor core, peritumoral edema, GD-enhancing tumor) and their potential clinical implications.
 4. **Clinical Implications:** Discuss how the insights gained from the segmentation results could guide clinical decision-making, including potential treatment strategies and areas for further investigation.
- While discussing the confidence of the segmentation and the model used, keep the explanations high-level and avoid overly technical details. The goal is to enhance the explainability of the AI segmentation to support clinical decision-making in a manner that can be easily understood by medical professionals.

Fig. 4. Prompt used for report generation.

Table 1

Computational details.

Component	Model
CPU	Intel(R) Xeon(R) Gold 5317, @ 3.00 GHz, 12 cores
GPU	NVIDIA A100 PCIe 80 GB
RAM	90 GB

curated mix of web documents, code, and dialogue data. Based on the Gemini research, it features a decoder-only Transformer architecture with multi-query attention optimized for efficient inference and safe conversational use.

- Llama 3 70B-IT [29]: It is the 70 billion parameter instruction-tuned version of Meta AI's LLM, which overcomes previous versions through innovations like grouped query attention, improved tokenization, and extensive curation of the 15 trillion token pre-training dataset across languages and domains.
- Mistral Saba 24B [30]: It is the 24 billion parameter instruction-tuned LLM from Mistral AI, designed for high-quality multilingual reasoning and tool use. Trained on a sophisticated data mix that includes synthetic tools and multilingual tasks, it builds on Mistral's dense decoder-only architecture with optimized attention mechanisms, yielding strong performance across benchmarks.

After careful prompt engineering, we designed an articulated prompt to provide the LLMs with context and guidance, enabling them to generate coherent and informative medical reports. The prompt incorporated the structured data from the JSON file, allowing the LLMs to integrate the quantitative information and generate human-readable textual descriptions of the tumor location, extent, and potential impact on brain functions. Integrating structured data into the prompt helps mitigate the risk of hallucinations and ensures accurate and relevant medical reports are generated. Fig. 4 shows the prompt used for the brain tumor task.

The following section details the experimental setup and evaluation results obtained by applying the proposed framework to brain MRI datasets for glioma and MS lesion segmentation.

4. Experimental results

4.1. Setting

We used a robust hardware setup for the experiments, as detailed in Table 1, and several specialized libraries: PyTorch [31] as the deep learning framework; MONAI [32] for implementing semantic segmentation models, data preprocessing, and metric calculations; Nibabel [33] and Nilearn [34] for reading, manipulating, and visualizing MRI data; and Siibra [35] for all operations related to brain atlas mapping. The

Table 2

Segmentation model hyperparameters.

Hyperparameter	Value
Optimizer	Adam
Learning rate	1e-4
Weight decay	1e-5
Loss	Dice loss
Scheduler	Cosine Annealing LR
Epochs	100
Patience	10

Table 3

LLM hyperparameters.

Hyperparameter	Value
Decoding strategy	Greedy
Temperature	0.6
Max tokens	1024
Top p	0.9

Groq API [36] was used to access and interact with the LLMs for the textual explainability component. A comprehensive overview of the critical hyperparameters used during the segmentation model training phase and the LLM prompting phase is provided in Tables 2 and 3, respectively. All other parameters not listed follow the default settings of the corresponding libraries.

4.1.1. Semantic segmentation

We employed the following experimental setup. The dataset was divided into 80% for training and 20% for testing, with an additional 20% of the training set used for validation.

We employed state-of-the-art deep learning models for semantic segmentation and assessed their performance using task-specific metrics for brain tumor segmentation, computed across the entire test set and complemented by 95% confidence intervals estimated through bootstrapping with 1000 iterations to capture statistical variability. The performance of the employed models was assessed using two widely adopted metrics:

- Dice Score: It measures the similarity between the predicted segmentation masks and the ground truth masks, ranging from 0 to 1, with higher values indicating better segmentation accuracy. The Dice Score is defined as:

$$\text{Dice} = \frac{2|A \cap B|}{|A| + |B|}, \quad (2)$$

where A and B are the predicted and ground truth binary masks, respectively.

- Hausdorff Distance: This metric quantifies the maximum distance between the predicted segmentation masks and the ground truth

Table 4

Dice scores for the three segmentation models: Residual U-Net, SegResNet, and SwinUNETR. The scores are reported for enhancing tumor (ET), tumor core (TC), whole tumor (WT), and the average score across these classes. For each metric, the corresponding 95% confidence interval (CI) is also provided.

Model	ET [CI 95%]	TC [CI 95%]	WT [CI 95%]	Average [CI 95%]
Residual U-Net	0.8323 [0.8074, 0.8558]	0.8878 [0.8673, 0.9048]	0.9109 [0.9012, 0.9204]	0.8770 [0.8615, 0.8908]
SegResNet	0.8585 [0.8353, 0.8797]	0.9130 [0.8955, 0.9282]	0.9289 [0.9208, 0.9364]	0.9001 [0.8867, 0.9119]
SwinUNETR	0.8511 [0.8262, 0.8740]	0.8992 [0.8802, 0.9164]	0.9217 [0.9125, 0.9305]	0.8907 [0.8758, 0.9049]

masks, with lower values indicating better boundary adherence. The Hausdorff Distance is defined as:

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} d(a, b), \sup_{b \in B} \inf_{a \in A} d(a, b) \right\}, \quad (3)$$

where $d(a, b)$ is the Euclidean distance between points a and b , and A and B are the sets of points on the boundaries of the predicted and ground truth masks, respectively.

4.1.2. Textual explainability

To assess the quality and effectiveness of the generated medical reports, we developed a comprehensive evaluation framework that included various metrics: lexical diversity, readability, coherence, and coverage of information. To support the calculation of these metrics, we employed the Spacy library [37]. Specifically, the following metrics were used to evaluate the output of LLMs:

:

- *Lexical diversity:*

- Type-Token Ratio (TTR): This measure of lexical diversity is defined as the ratio of unique words (types) to the total number of words (tokens) in a text [38]. A higher TTR value indicates greater lexical richness and diversity in the output. The formula for calculating TTR is:

$$\text{TTR} = \frac{\text{Number of Unique Words}}{\text{Total Number of Words}}. \quad (4)$$

- Maas' Index: This adaptation of the Type-Token Ratio aligns the measure with a logarithmic scale [38]. A lower score on Maas' Index signifies greater diversity in vocabulary. The index is calculated using the formula:

$$\text{Maas'} = \frac{\log(\text{nTokens}) - \log(\text{nTypes})}{\log(\text{nTokens})^2}. \quad (5)$$

- *Readability:*

- Flesch Reading Ease Score (FRES): The score evaluates the ease of reading a given text based on factors such as word length and sentence complexity [39]. The formula for calculating FRES is:

$$\text{FRES} = 206.835 - (1.015 \times \text{ASL}) - (84.6 \times \text{ASW}), \quad (6)$$

where ASL represents the Average Sentence Length, and ASW represents the Average Number of Syllables per Word.

- *Coherence:*

- Coherence Score (CohS): This newly proposed metric quantifies the coherence and flow of the generated textual explanations, assessing the logical structure and transitions between ideas within the output. The coherence score is calculated as follows:

$$\text{CohS} = \frac{1}{N-1} \sum_{i=1}^{N-1} \cos(S_i, S_{i+1}), \quad (7)$$

where N is the number of sentences and $\cos(S_i, S_{i+1})$ is the cosine similarity between the embeddings of sentences i and $i+1$.

- *Coverage of Information:*

- Embedding-based Coverage Score (ECS): This new metric, introduced in this paper, measures the coverage of information in the LLM's output by comparing the embeddings of the sentences in the prompt and the corresponding output. The coverage score is calculated as follows:

$$\text{ECS} = \frac{1}{N} \sum_{i=1}^N \cos(S_{\text{text}}, S_{\text{ref}_i}), \quad (8)$$

where:

- * S_{text} is the embedding of the LLM output.
- * S_{ref_i} is the embedding of the i th sentence in the reference text (LLM prompt).
- * N is the total number of sentences in the reference text.
- * $\cos(S_{\text{text}}, S_{\text{ref}_i})$ is the cosine similarity between the embedding of the LLM output and the reference text (prompt) sentence.

- Token-based Coverage Score (TCS): This newly proposed metric quantifies the overlap between the tokens in the prompt and the output. It is calculated as the ratio of the intersection of tokens to their union after removing punctuation and stop words from both sets:

$$\text{TCS} = \frac{|\text{textTokens} \cup \text{refTokens}|}{|\text{textTokens} \cap \text{refTokens}|}. \quad (9)$$

This metric evaluates how well the LLM incorporated relevant information from the prompt into its generated output.

4.2. Results

4.2.1. Semantic segmentation

The Dice Scores obtained for the three segmentation models (Residual U-Net, SegResNet, and SwinUNETR) are presented in Table 4. These scores reflect the performance of each model for the three segmentation classes, enhancing tumor (ET), tumor core (TC), and whole tumor (WT), and their averages. The Hausdorff Distances for the three models are shown in Table 5. These distances indicate the boundary adherence of each model for the three segmentation classes and their averages.

Based on the quantitative results, the SegResNet model emerged as the best-performing architecture for the brain tumor segmentation task, achieving the highest values for the performance metrics across all classes. The SwinUNETR model demonstrated competitive performance, while the U-Net model lagged in both evaluation metrics.

Given these results, the SegResNet model was selected for use in the subsequent stages of the pipeline. Examples of the segmentation provided by SegResNet for brain tumors are shown in Fig. 5.

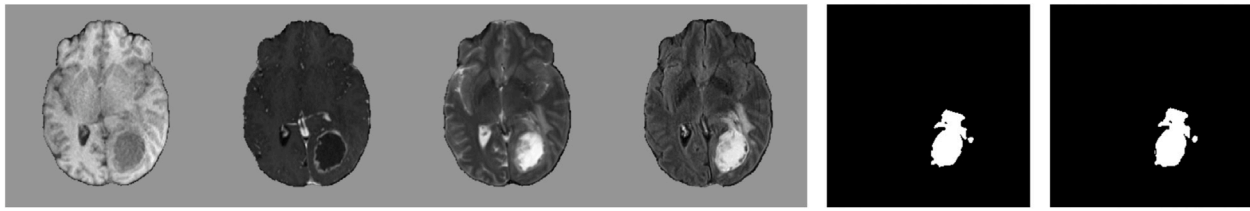


Fig. 5. Results provided by the SegResNet model for brain tumor segmentation. From left to right: 4-channel MRI input scan, ground truth mask for the whole tumor (WT), and predicted segmentation mask.

Table 5

Hausdorff Distances for the three segmentation models: Residual U-Net, SegResNet, and SwinUNETR. The distances are reported for enhancing tumor (ET), tumor core (TC), whole tumor (WT), and the average distance across these classes.

Model	ET	TC	WT	Average
Residual U-Net	4.9207	5.4873	8.0632	6.1571
SegResNet	3.4936	4.0667	4.5787	4.0463
SwinUNETR	4.0749	4.7311	7.6993	5.5018

4.2.2. Textual explainability

The LLMs employed in our framework – Gemma, Llama, and Mistral – exhibited varying performance levels in generating textual explanations for the segmentation task. Each model demonstrated unique strengths across different aspects of text generation. We evaluated the performance of these LLMs using several metrics: lexical diversity, readability, coherence, and coverage of information. We recognize the critical importance of refining prompt engineering to ensure consistent and reliable report generation across different LLMs. To address this concern, we emphasize that an empirical selection process was conducted, prioritizing a prompt design that optimizes coherence and comprehensive coverage while minimizing inter-model performance differences. Table 6 summarizes the evaluation results for the brain tumor segmentation task. For each LLM, we report the average performance across all generated reports in the test set, along with the standard deviation of each metric, to provide insight into the consistency of the outputs.

The results in Table 6 highlight the diverse strengths among the models. Gemma demonstrated superior lexical richness, achieving the highest Type-Token Ratio of 0.351 and the lowest Maas' Index of 0.023, indicating a rich and diverse vocabulary. Mistral led in coherence with a score of 0.357. Llama slightly outperformed the other models in readability with a Flesch Reading Ease Score of 37.204, Embedding-based Coverage (0.433) and Token-based Coverage (0.305). An example of the output generated by Llama for the brain tumor segmentation is shown in Fig. 6.

4.2.3. Error evaluation

Although this approach offers substantial benefits, it has shortcomings and could be further refined. For instance, our top-performing model achieves an average Dice Score of 0.90 for segmentation mask generation in the brain tumor task. Nevertheless, even minor inaccuracies in these masks can lead to ambiguous visual explanations. These imprecisions can also cascade into subsequent analyses, potentially leading to misinterpretations of the tumor's dimensions, shape, spread, and potential effects on brain functionality. The model's segmentation errors, whether over-segmentation (excessive marking of tumor area) or under-segmentation (omitting tumor parts) are distributed across the entire tumor region. In our framework, we introduced a specific mitigation strategy to reduce the impact of such errors on the report generation process. By deliberately restricting the input to the five brain areas most significantly affected by the disease, the pipeline limits the propagation of false positives and false negatives to the textual

output. This targeted filtering is a purposeful design choice within our framework and should not be considered an inherent property of all segmentation-to-report pipelines. This constraint, while limiting, safeguards against propagating segmentation errors.

To quantitatively assess the impact of segmentation inaccuracies and further validate the applied mitigation strategy, we conducted a comparative analysis across the entire test set. Specifically, we examined the brain regions extracted using the segmentation masks generated by the SegResNet model against their corresponding ground truth. We then progressively reduced the number of regions considered for medical report generation, starting from a maximum of 30 (the highest value observed in the test set) down to a single region.

The results, presented in Fig. 7, show the percentage of test set examples containing at least one region that would not have been affected by the tumor if identified using the corresponding ground truth. Notably, by setting a threshold of five regions for medical report generation, we achieve a 96.6% reduction in error propagation, decreasing from 53.1% to 1.8% for cases with at least one incorrect region out of five. Furthermore, for cases with two incorrect regions out of five, the error is reduced by 98.3%, from 23.5% to 0.4%. Importantly, no instances were found with more than two erroneous regions out of five, demonstrating the high reliability of our error mitigation strategy and the overall robustness of our methodology in minimizing segmentation inaccuracies and ensuring consistency in medical report generation.

4.2.4. Statistical analysis

To determine whether observed differences in report quality are statistically meaningful across the evaluated LLMs (Gemma, Llama, and Mistral) we conducted pairwise statistical comparisons on the six text quality metrics: Type-Token Ratio (TTR), Maas' Index, Flesch Reading Ease Score (FRES), Coherence Score (CohS), Embedding-based Coverage Score (ECS), and Token-based Coverage Score (TCS). Given that each LLM generated reports for the same test set instances, we employed paired statistical tests, ensuring that comparisons were made on a per-sample basis.

For each metric, the choice of statistical test was guided by the distribution of paired differences. When the normality assumption held, verified using the Shapiro–Wilk test [40], we applied a paired t-test; otherwise, we used the Wilcoxon signed-rank test, a non-parametric alternative suitable for skewed or ordinal data.

The results of the pairwise comparisons are summarized in Table 7. Significant differences (with $p < 0.05$) were found between Llama and Mistral for all six metrics. Llama outperformed Mistral in vocabulary diversity (TTR and Maas'), readability (FRES), coherence (CohS), and both coverage metrics (ECS and TCS), with all p-values indicating strong evidence of difference. Comparisons between Llama and Gemma also revealed significant differences across all metrics. These results indicate that Llama consistently produced outputs with statistically distinct linguistic and semantic characteristics compared to Gemma, suggesting meaningful performance differentiation between the two models. In contrast, the comparison between Mistral and Gemma yielded non-significant differences for Coherence Score ($p =$

Table 6

Performance metrics for brain tumor segmentation across different LLMs. Metrics include Type-Token Ratio (TTR), Maas' Index (Maas'), Flesch Reading Ease Score (FRES), Coherence Score (CohS), Embedding-based Coverage Score (ECS), and Token-based Coverage Score (TCS). For each metric, the mean and standard deviation across the test set are reported to reflect the variability in the LLM outputs.

Metric	Definition	Gemma	Llama	Mistral
TTR	The ratio of unique words (types) to the total number of words (tokens) in a text.	0.351 ± 0.019	0.282 ± 0.031	0.276 ± 0.024
Maas'	TTR on a logarithmic scale.	0.023 ± 0.021	0.029 ± 0.002	0.027 ± 0.002
FRES	The ease of reading a given text based on factors such as word length and sentence complexity.	32.224 ± 4.749	37.204 ± 5.359	34.108 ± 6.059
CohS	The coherence of the generated text, assessing the logical structure and transitions between ideas within the output.	0.357 ± 0.039	0.319 ± 0.071	0.357 ± 0.036
ECS	The coverage of information by comparing the embeddings of the sentences in the prompt and the corresponding output.	0.426 ± 0.016	0.433 ± 0.020	0.424 ± 0.019
TCS	The coverage of information by comparing the overlap between the tokens in the prompt and the output.	0.257 ± 0.022	0.305 ± 0.033	0.278 ± 0.024

The brain tumor segmentation results indicate that the tumor is primarily located in the right hemisphere of the brain, with a significant presence in the Temporal-to-Parietal (GapMap) region (49.22% of the tumor volume). The tumor also affects the Area hOc4la (LOC), Area hOc4lp (LOC), Area hIP4 (IPS), and Area PGp (IPL) regions, albeit to a lesser extent. The semantic segmentation of the tumor reveals three distinct categories: tumor core (red), peritumoral edema (yellow), and GD-enhancing tumor (green). These categories have important implications for understanding the tumor's behavior and potential treatment strategies.

Detailed Regional Impact

- 1. Temporal-to-Parietal (GapMap) right:** The tumor occupies 15.55% of this region, which is responsible for processing auditory and visual information. The presence of the tumor may impact the patient's ability to process sensory information, leading to potential cognitive and perceptual deficits.
- 2. Area hOc4la (LOC) right:** The tumor affects 89.44% of this region, which is involved in object recognition and visual processing. The tumor's presence may lead to difficulties in recognizing objects and understanding visual stimuli.
- 3. Area hOc4lp (LOC) right:** The tumor occupies 84.2% of this region, which is also involved in object recognition and visual processing. Similar to the previous region, the tumor's presence may impact the patient's ability to recognize objects and understand visual information.
- 4. Area hIP4 (IPS) right:** The tumor affects 97.44% of this region, which is responsible for processing spatial information and attention. The tumor's presence may lead to difficulties in spatial awareness, attention, and executive functions.
- 5. Area PGp (IPL) right:** The tumor occupies 22.97% of this region, which is involved in language processing and semantic memory. The tumor's presence may impact the patient's language abilities and memory.

Semantic Segmentation

The semantic segmentation of the tumor reveals three distinct categories:

- 1. Tumor Core (red):** This region represents the most aggressive and malignant part of the tumor, which requires immediate attention and treatment.
- 2. Peritumoral Edema (yellow):** This region represents the area surrounding the tumor that is affected by the tumor's growth and invasion. It may indicate the tumor's potential to spread and invade surrounding brain tissue.
- 3. GD-Enhancing Tumor (green):** This region represents the part of the tumor that is most responsive to contrast agents, indicating active tumor growth and potential treatment targets.

Clinical Implications

The insights gained from the segmentation results can guide clinical decision-making in several ways:

- 1. Surgical planning:** The detailed regional impact and semantic segmentation can help surgeons identify the most critical areas to target during tumor resection.
- 2. Treatment strategies:** The presence of peritumoral edema and GD-enhancing tumor may indicate the need for adjuvant therapies, such as chemotherapy or radiation therapy, to target these areas.
- 3. Monitoring and follow-up:** The segmentation results can help clinicians monitor the tumor's growth and response to treatment, allowing for timely adjustments to the treatment plan.
- 4. Further investigation:** The affected regions and semantic segmentation categories can guide further investigation into the tumor's molecular biology and potential biomarkers for targeted therapies.

The high confidence of the segmentation results and the use of the SegResNet model ensure that the results are reliable and accurate, providing a solid foundation for clinical decision-making.

Fig. 6. Example of Llama output for the brain tumor segmentation task.

0.97) and Embedding-based Coverage Score ($p = 0.07$). This suggests that, for these two metrics, Mistral and Gemma produced broadly similar outputs. However, significant differences were still observed in TTR, Maas', FRES, and TCS, indicating that Mistral and Gemma vary in terms of lexical richness and surface-level textual properties.

These statistical results quantitatively confirm that the three LLMs exhibit distinct strengths and weaknesses across different aspects of report quality. In particular, Llama demonstrates consistent superiority in both syntactic and semantic dimensions, while Mistral and Gemma show overlap in certain areas but diverge in others. The use of paired statistical tests guarantees that the observed differences are derived from robust, sample-wise comparisons. These findings indicate that the

LLMs exhibit systematic, rather than random, variations in performance across metrics, offering a statistically grounded basis for interpreting and comparing their behavior in clinical text generation tasks.

4.3. Beyond brain tumors

The results' applicability can be extended far beyond the specific domain of brain tumor segmentation. The framework can be adapted and generalized to various medical imaging modalities and pathologies, thereby encouraging the more widespread adoption of explainable AI systems in healthcare. We applied the proposed methodology to a second clinical scenario: segmenting multiple sclerosis (MS) lesions.

Table 7

Statistical comparison of text quality metrics between LLM-generated reports. Pairwise comparisons were conducted between Gemma, Llama, and Mistral using either the paired t-test (T) or the Wilcoxon signed-rank test (W), depending on the distribution of the differences. Metrics evaluated include Type-Token Ratio (TTR), Maas' Index, Flesch Reading Ease Score (FRES), Coherence Score (CohS), Embedding-based Coverage Score (ECS), and Token-based Coverage Score (TCS). For each comparison, the statistical test used, p -value, and significance (at $p < 0.05$) are reported.

Metric	Llama-Mistral			Llama-Gemma			Mistral-Gemma		
	Test	p-value	Significant	Test	p-value	Significant	Test	p-value	Significant
TTR	W	0.03	True	W	<0.001	True	T	<0.001	True
Maas'	T	<0.001	True	W	<0.001	True	W	<0.001	True
FRES	W	<0.001	True	W	<0.001	True	W	<0.001	True
CohS	W	<0.001	True	W	<0.001	True	T	0.97	False
ECS	T	<0.001	True	T	<0.001	True	T	0.07	False
TCS	T	<0.001	True	T	<0.001	True	T	<0.001	True

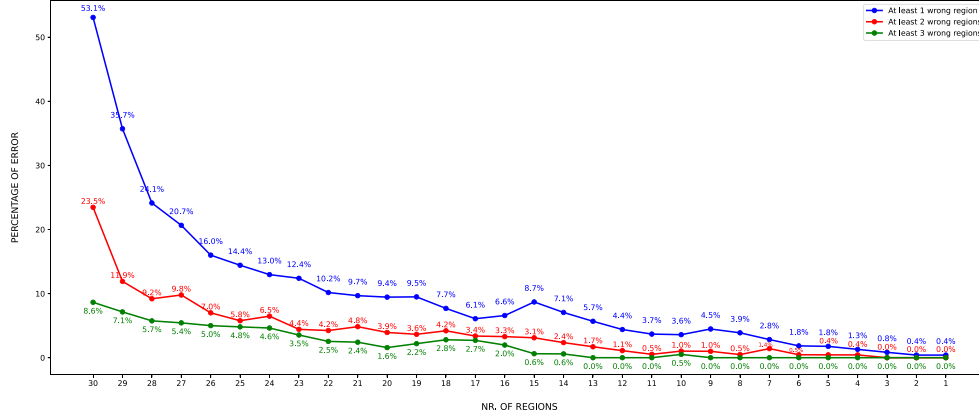


Fig. 7. Percentages of error reduction achieved through the proposed mitigation strategy. The chart reports the reduction in region-level mismatches across test set cases, grouped by severity: cases with at least one, at least two, and at least three mismatched regions between the predicted and ground truth annotations.

4.3.1. Data

For the MS lesion segmentation task, a separate dataset was used, comprising three distinct subsets: MSSEG-1 [41], ISBI [42], and PubMRI [43]. This dataset is part of the 2022 Shifts challenge [44] and focuses on white matter MS lesion segmentation in 3D MRI of the brain. Each sample in the MS dataset consists of a 3D brain scan captured using both FLAIR and T1 contrasts. These scans have undergone several preprocessing steps to enhance their quality and consistency, including denoising to reduce noise and improve image clarity, skull stripping to remove non-brain tissues and focus the analysis on brain matter, and bias field correction to address intensity non-uniformities in the images. The combination of FLAIR and T1 provides complementary information, with FLAIR being particularly sensitive to white matter lesions and T1 offering detailed anatomical structure.

4.3.2. Method adaptations

The methodology is the same as that applied for brain tumors, described in Section 3, with a few adaptations to make it suitable for MS lesions.

A similar JSON file was constructed for the MS lesion segmentation task. The main difference from the brain tumor JSON file is the absence of color coding. Unlike the brain tumor JSON, which includes color information for different lesion types, the MS lesion JSON does not, as MS lesions are typically represented as a single class.

For the MS lesion segmentation task, the performance was evaluated using the metrics suggested by the Shifts challenge:

- Normalized Dice Similarity Coefficient (nDSC) [45]: It provides an unbiased metric for evaluating model performance in identifying lesions, regardless of the lesion load across different patients. It normalizes the precision for each patient while keeping the recall fixed, allowing for fair comparison across different patients. The nDSC is defined as:

$$\text{nDSC} = 2(2 + \kappa p + n)^{-1}, \quad \kappa = h(r^{-1} - 1), \quad (10)$$

where $p = \frac{FP}{TP}$ and $n = \frac{FN}{TP}$. In this context, h represents the ratio of the positive to the negative classes in the predicted mask $\hat{Y}$, and $0 < r < 1$ is a reference value that denotes the mean fraction of the positive class in $\hat{Y}$ across a large number of subjects, indicating the average occurrence rate of the positive class.

- F1 Lesion Score: It assesses the lesion detection quality and is defined as:

$$F_1 = \frac{TP}{TP + 0.5(FP + FN)}, \quad (11)$$

where TP (true positives) is when the detected lesion matches the actual lesion well (IoU > 0.5), FP (false positives) is when the predicted lesion does not match any actual lesion closely (IoU < 0.5), and FN (false negatives) is when an actual lesion is not detected correctly (IoU < 0.5).

- Robust Area Under the Curve (R-AUC) [44,46]: This metric jointly assesses the model's robustness to distributional shifts and the quality of its uncertainty estimates. It is calculated as the area under error-retention curves, plotting the mean error against the fraction of retained predictions. Lower R-AUC indicates better performance, achieved by improving prediction accuracy or better correlating uncertainty estimates with errors.

4.3.3. Results

The nDSC, F1 Lesion Score, and R-AUC results for the three segmentation models (Residual U-Net, SegResNet, and SwinUNETR) on the MS lesion segmentation task are presented in Table 8. Consistent with the brain tumor segmentation results, the SegResNet model outperformed the other architectures in the MS lesion segmentation task, achieving the highest scores across all three metrics. It was selected for use in the subsequent stages of the pipeline. Examples of the segmentation provided by SegResNet for MS lesions are shown in Fig. 8.

We also evaluated the performance of LLMs using the same metrics described above for the brain tumor segmentation task. The results

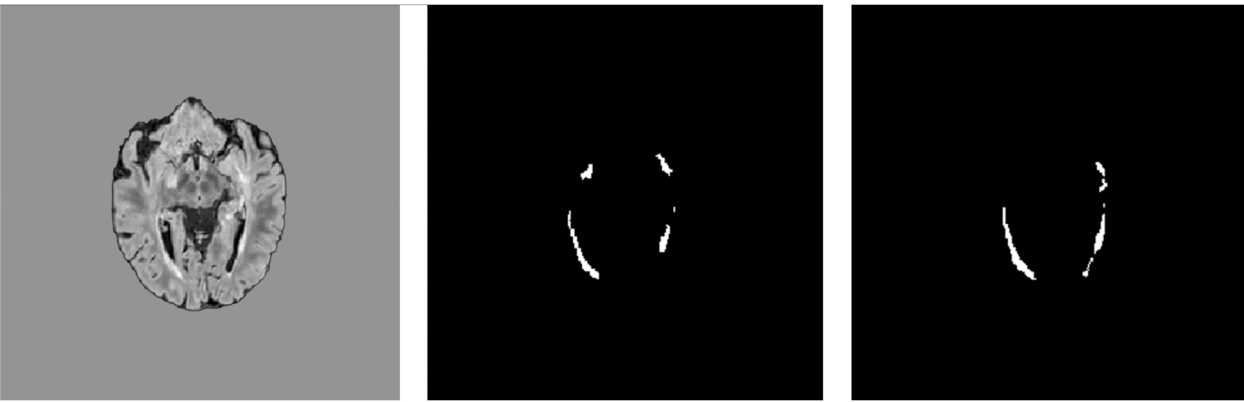


Fig. 8. Results provided by the SegResNet model for multiple sclerosis lesion segmentation. From left to right: FLAIR MRI input scan, ground truth mask, and predicted segmentation mask.

Table 8

Performance metrics for multiple sclerosis lesion segmentation using the three segmentation models: Residual U-Net, SegResNet, and SwinUNETR. Metrics include Normalized Dice Similarity Coefficient (nDSC), F1 Lesion Score (F1), and Robust Area Under the Curve (R-AUC).

Model	nDSC	F1	R-AUC
Residual U-Net	0.6505	0.2094	0.1764
SegResNet	0.7175	0.3511	0.8745
SwinUNETR	0.6755	0.3263	0.3529

Table 9

Performance metrics for multiple sclerosis lesion segmentation across different LLMs. Metrics include Type-Token Ratio (TTR), Maas' Index (Maas'), Flesch Reading Ease Score (FRES), Coherence Score (CohS), Embedding-based Coverage Score (ECS), and Token-based Coverage Score (TCS).

Model	TTR	Maas'	FRES	CohS	ECS	TCS
Gemma	0.533	0.019	17.600	0.385	0.595	0.230
Llama	0.381	0.026	41.770	0.354	0.541	0.295
Mistral	0.453	0.020	41.560	0.334	0.559	0.268

varied for the MS lesion segmentation task, as shown in Table 9. Gemma achieved the highest lexical richness with a TTR of 0.533 and the lowest Maas' Index of 0.019, again demonstrating superior vocabulary diversity. Regarding readability, Llama led with an FRES of 41.77, closely followed by Mistral at 41.56. Gemma maintained its strong performance in Coherence with a score of 0.385 and excelled in Embedding-based Coverage (0.595). However, Llama achieved the best Token-based Coverage (0.295). An example of the output generated by Llama for the MS lesion segmentation task is shown in Fig. 9.

These results establish a solid foundation for further analysis, which we will discuss in the following section regarding methodological implications, limitations, and potential future directions.

5. Discussion and conclusion

This paper presented an innovative framework to enhance the explainability of AI systems in medical imaging, specifically for brain segmentation tasks. By integrating state-of-the-art semantic segmentation techniques, brain atlas mapping, and Large Language Models, our approach generates comprehensive, clinically meaningful textual reports that align with physicians' diagnostic reasoning and reporting practices. Unlike conventional explainability methods that rely on heatmaps or concept attribution, our framework offers structured, anatomically grounded, and linguistically validated explanations – marking a step towards clinically deployable AI transparency.

A distinctive aspect of the framework is its anti-hallucination design, which incorporates structured data extraction and a JSON-based intermediary combined with constrained prompts to ground LLM outputs in factual, quantitative information. This approach addresses a common limitation in naïve LLM-based report generation, significantly reducing the risk of hallucinated content. Furthermore, by bridging visual and textual modalities, the framework provides a more holistic explanation, improving the interpretability of segmentation outputs for medical professionals.

To ensure robustness and flexibility, we conducted multi-model validation using different segmentation models and multiple LLMs, demonstrating consistent performance across architectures and confirming the model-agnostic applicability of our method. The SegResNet model emerged as the top performer for semantic segmentation, achieving high Dice Scores and low Hausdorff Distances for brain tumor segmentation, as well as competitive performance in MS lesion segmentation. Using a comprehensive set of metrics, we evaluated three state-of-the-art LLMs (Gemma, Llama, and Mistral) in the textual explanation generation phase. The results revealed that each model had unique strengths in different aspects of text generation, such as lexical diversity, readability, coherence, and information coverage. Llama consistently performed well across both tasks, excelling in lexical diversity and coverage metrics.

While our framework was primarily implemented for segmenting brain tumors in magnetic resonance imaging of the brain, its versatility allows for broader applications. The underlying principles of this approach can be easily adapted and applied to the segmentation of various other medical conditions across different imaging modalities, as demonstrated by the MS lesion detection results. This work contributes to the ongoing efforts to make AI systems in healthcare more transparent, interpretable, and trustworthy. By providing medical professionals with detailed, context-rich explanations of AI-driven analyses, our approach has the potential to support clinical decision-making and increase confidence in AI-assisted diagnoses.

While this approach offers significant benefits, it has limitations that warrant further refinement. Our top-performing model achieves an average Dice Score of 0.90 in brain tumor segmentation, yet minor inaccuracies can lead to ambiguous visual explanations and misinterpretations of tumor characteristics. Segmentation errors, whether over or under-segmentation, are distributed across the tumor region. To mitigate their impact on report generation, our framework includes a dedicated strategy that limits the input to the five brain areas most affected by the disease, reducing error propagation by up to 98.3%. Additionally, to further enhance the transparency and interpretability of our decision-making process, we incorporate segmentation confidence values into the JSON construction. These values are subsequently

The AI model has identified multiple sclerosis (MS) lesions in various regions of the brain according to the Julich-Brain Atlas. The lesions are predominantly located in the frontal and occipital areas, with a significant presence in the left hemisphere. Notably, the Frontal-to-Occipital (GapMap) left and right regions are affected, suggesting potential disruptions to the communication between the frontal and occipital lobes. The presence of lesions in the Area hOc1 (V1, 17, CalcS) left and Area hOc6 (POS) left regions may impact visual processing and object recognition.

Detailed Regional Impact

- 1. Frontal-to-Occipital (GapMap) left:** The lesion occupies 26.88% of the region, with 6.04% of the region affected. This region is involved in integrating information from the frontal and occipital lobes, which may be disrupted due to the lesion presence. Functions potentially impacted include attention, memory, and visual processing.
- 2. Frontal-to-Occipital (GapMap) right:** The lesion occupies 16.82% of the region, with 4.08% of the region affected. Similar to the left hemisphere, the lesion presence may disrupt communication between the frontal and occipital lobes, affecting attention, memory, and visual processing.
- 3. Area hOc1 (V1, 17, CalcS) left:** The lesion occupies 8.88% of the region, with 7.31% of the region affected. This region is primarily responsible for primary visual processing, including edge detection and line orientation. The lesion presence may impact visual acuity and processing.
- 4. Area hOc6 (POS) left:** The lesion occupies 4.23% of the region, with 51.55% of the region affected. This region is involved in object recognition and visual processing. The significant lesion presence may lead to difficulties in object identification and recognition.
- 5. Frontal-to-Temporal-II (GapMap) left:** The lesion occupies 3.1% of the region, with 2.86% of the region affected. This region is involved in language processing and semantic memory. The lesion presence may impact language comprehension and production.

Clinical Implications

The semantic segmentation results highlight the presence of MS lesions in various regions of the brain, which may impact cognitive and motor functions. The lesions' presence in the frontal and occipital areas may lead to:

1. Disruptions in attention, memory, and visual processing due to the lesions in the Frontal-to-Occipital (GapMap) regions.
2. Impaired visual acuity and processing due to the lesion in the Area hOc1 (V1, 17, CalcS) left region.
3. Difficulty with object recognition and visual processing due to the lesion in the Area hOc6 (POS) left region.
4. Potential language comprehension and production difficulties due to the lesion in the Frontal-to-Temporal-II (GapMap) left region.

The AI model's segmentation results provide valuable insights into the location and impact of MS lesions on brain function. These findings can support clinical decision-making by identifying areas of potential cognitive and motor impairment, enabling clinicians to develop targeted treatment plans and monitor disease progression.

Fig. 9. Example of Llama output for the MS lesion segmentation task.

reflected in the generated medical report, providing an additional layer of factual basis and allowing for a more informed interpretation of the results. The strategy of focusing on the top five most affected brain regions has proven effective in mitigating error propagation; however, it may miss smaller yet clinically important findings, and its reliability depends on the accuracy of atlas registration, which can be influenced by anatomical variability, surgical alterations, or pathological deformations. Given these considerations, we acknowledge the potential benefits of dynamic thresholding and uncertainty calibration and consider them promising directions for future research.

Moreover, the quality of the generated reports remains sensitive to prompt design and model-specific characteristics. Despite using consistent input structures, different LLMs exhibit varied behavior in terms of coherence, coverage, and phrasing. This highlights the need for further optimization of prompt strategies and model selection, particularly for clinical deployment where consistency and reliability are critical.

Although the framework has demonstrated consistent performance on well-curated datasets such as BraTS and Shifts, real-world clinical settings may introduce variability in scanner types, imaging protocols, and patient populations. While not directly addressed in this study, managing potential domain shifts remains an important consideration for future work to guarantee broad applicability across diverse healthcare environments.

Additional future work could focus on refining LLM prompts by integrating patient metadata and medical history for more personalized reports, embedding the system into clinical workflows for effective patient monitoring, and adapting the methodology to other imaging tasks like lesion detection and disease classification. While our evaluation relies on quantitative text-based metrics (e.g., readability, coherence, lexical coverage), it is essential to complement these with clinical user studies to assess report usefulness and real-world applicability. Additional imaging techniques and patient information can improve report accuracy, and implementing interactive feedback systems can help refine the process. Creating domain-specific language models and conducting clinical validation studies with medical professionals will ensure the system's performance, usability, and alignment with real-world medical practices.

CRediT authorship contribution statement

Alberto G. Valerio: Writing – review & editing, Writing – original draft, Software, Methodology. **Katya Trufanova:** Software, Methodology. **Salvatore de Benedictis:** Software. **Gennaro Vessio:** Writing – review & editing, Validation, Supervision, Methodology. **Giovanna Castellano:** Validation, Supervision, Methodology.

Funding information

One author, Giovanna Castellano, has received funding from the FAIR – Future AI Research project, Spoke 6 – Symbiotic AI (Grant number: CUP H97G22000210007), under the NRRP MUR program funded by NextGenerationEU. None of the other authors received any specific funding.

Declaration of competing interest

There are no competing interests associated with this manuscript. None of the authors have any financial, personal, or professional interests that could be perceived to influence the work reported in this study.

Acknowledgments

While preparing this work, the authors utilized ChatGPT and Grammarly to enhance language clarity and readability. The authors, who take full responsibility for the final version of the manuscript, carefully reviewed and refined all content generated by these tools.

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